

Uncovering Chronic Kidney Disease

A Data Driven Approach to Diagnosis and Risk Discovery

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1. Introduction

For our analysis, we used the Chronic Kidney Disease dataset from the UC Irvine Machine Learning Repository. Chronic kidney disease is a serious and common health condition that involves the gradual loss of kidney function over time. In fact, “more than 1 in 7 adults in the US, or about 37 million people, may have chronic kidney disease (CKD)” (*High blood pressure and chronic kidney disease*). Since the kidneys are responsible for filtering waste and excess fluids from our blood, any damage to the kidneys can lead to severe health complications. If left untreated, chronic kidney disease can eventually progress to kidney failure, requiring dialysis or even a kidney transplant (*Chronic Kidney Disease*).

There are many factors that can increase a patient’s risk of developing chronic kidney disease. According to the National Institute of Diabetes and Digestive and Kidney Diseases, the two most common causes of chronic kidney disease in adults are diabetes and high blood pressure (Chronic Kidney Disease (CKD) - NIDDK). These risk factors are directly reflected in the predictors provided in our dataset and played an important role in performing our analysis and understanding our results.

The dataset itself contains a mix of clinical and laboratory predictors commonly used in medical evaluations. Clinical features include a patient’s age, blood pressure, or history of conditions like diabetes or hypertension. These are typically recorded during routine medical evaluations. Laboratory features like blood urea (bu), serum creatinine (sc), and urine specific gravity (sg) are collected through blood and urine tests. These test measures help assess kidney function and detect any abnormalities. In a medical setting, this information provides doctors with a dynamic picture of a patient’s overall kidney health. In the dataset, these values are represented by numerical and categorical data. We will discuss a deeper analysis of the predictors in the *Relevance to the Predictors* section.

Research Questions

The first research question we addressed was whether we can predict whether a patient has chronic kidney disease based on clinical and laboratory features. To explore this, we used logistic regression and random forest classification to compare model performance and interpretability.

The second question we addressed was whether we can identify distinct subgroups of patients based on their clinical and laboratory features. We used various clustering algorithms to carry out this analysis, including K-Means clustering and hierarchical clustering.

The third question we explored was whether we can detect early signs of chronic kidney disease. To investigate this, we applied classification and regression trees (CART) and random forest algorithms to observe variable importance. This allows us to determine which features strongly

influence the prediction of chronic kidney disease and can provide insight into potential early signs.

Description of Dataset

This Chronic Kidney Disease dataset used in this analysis was obtained from the UC Irvine Machine Learning Repository. The data was originally donated by Dr. P. Soundarapandian, who is a nephrologist at Apollo Hospitals in Tamil Nadu, India. For reference, a nephrologist is a medical doctor specializing in the diagnosis and treatment of kidney diseases. The dataset was prepared by L. Jerlin Rubini who is a research scholar at Alagappa University and was supervised by Dr. P. Eswaran, Assistant Professor in the Department of Computer Science and Engineering at Alagappa University.

The dataset includes 400 patient records with 24 predictors and one response variable, class. The target variable class has two levels: ckd and notckd. A value of ckd indicates that the patient was diagnosed with chronic kidney disease, while notckd signifies the patient was not diagnosed with chronic kidney disease. The laboratory predictors in the dataset are likely derived from blood and urine tests. The clinical predictors reflect patient history and include age, patient blood pressure, diabetes mellitus, hypertension, and more. Together, these variables provide a comprehensive profile for assessing kidney function and related health conditions. To conduct a meaningful analysis, it is important to first understand the background of these predictors and how they relate to the diagnosis of chronic kidney disease.

Relevance of Predictors

The column ‘age’ refers to the patient’s age in years and is stored as an integer value. Individuals over the age of 60 are more likely to develop chronic kidney disease, and “more than 50 percent of seniors over the age of 75 are believed to have kidney disease” (*Aging and Kidney Disease*).

The ‘bp’ column corresponds to the patient’s blood pressure, specifically diastolic pressure. It is also stored as an integer value in mm/Hg units. This measures “the force of the blood when the heart is between beats.” Hypertension is another critical risk factor for chronic kidney disease, and uncontrolled high blood pressure is the second leading cause of kidney failure in the United States. Although our dataset is based on patients from Tamil Nadu, India, the global significance of hypertension is evident as “nearly half of all adults in the US (108 million or 45% of the population) have high blood pressure” (*High Blood Pressure and Chronic Kidney Disease*).

The ‘sg’ column refers to the specific gravity of urine, and this value can be collected from a standard urinalysis test. This value indicates how concentrated a patient's urine is. Ideally, specific gravity results should fall between 1.002 and 1.030 when kidneys are functioning normally (Nall). Abnormal values outside of this range may suggest kidney dysfunction or

chronic kidney disease. However, during our research, we found that specific gravity levels can vary significantly throughout the day depending on a person's hydration status. Because of this, a single measurement of urine specific gravity may not be the most reliable indicator for diagnosing chronic kidney disease. Instead, monitoring this measure over time allows for a more relevant assessment.

The 'al' column corresponds to albumin levels. Albumin is a protein found in the blood that plays a role in maintaining proper fluid balance in the body. Studies have shown that chronic kidney disease patients with low serum albumin levels have an increased risk of progressing to kidney failure. According to the National Kidney Foundation, the normal range for serum albumin is between 3.5 g/dL and 5 g/dL (*Kidney failure risk factor: Serum albumin*).

The 'su' column represents the sugar level in a patient's urine. This measure is closely related to diabetes, one of the leading causes of chronic kidney disease. Normally, urine should contain little to no sugar, as the kidneys should prevent glucose from being excreted. However, elevated urine sugar levels ($su \geq 2$) may indicate underlying or uncontrolled diabetes (Cleveland Clinic, Glycosuria: What it Means & When You Need Treatment).

The 'rbc' column represents the presence of red blood cells in a patient's urine. This condition, known as hematuria, can be a sign of damage to the filters in the kidneys. While small amounts of blood in urine may result from benign causes, persistent hematuria may suggest underlying kidney dysfunction (Mayo Clinic, Blood in Urine (Hematuria)).

The 'pc' column refers to pus cells, which indicate the presence of white blood cells in the urine. Elevated white blood cells (pyuria) may suggest a urinary tract infection or inflammation in the urinary system. Persistent pyuria can be associated with ongoing renal damage. Similarly, the 'pcc' column refers to pus cell clumps, which represent a more advanced stage of infection. The presence of clumps may suggest that the infection is long standing or has spread deeper into the kidney tissue (Dr Lal PathLabs, Pus Cells in Urine: Causes & Diagnosis).

The 'ba' column refers to the presence of bacteria in a patient's urine. This typically indicates a urinary tract infection (UTI). While UTIs are common and often treatable, recurrent infections can lead to kidney infections. If left untreated, these infections may cause permanent kidney damage over time (Dicu-Andreescu et al.).

The 'bgr' column represents the patient's random blood glucose level. This measures the concentration of glucose in the blood at any given time, regardless of whether the blood sample was taken when the patient was fasting or not. Diabetes is the leading cause of chronic kidney disease worldwide, so blood glucose is a critical variable in chronic kidney analysis. According to Mount Sinai studies, a random blood glucose level greater than 200 mg/dL is a strong indicator of diabetes. A normal value is typically below 140 mg/dL. In our dataset, we observed

that nearly all patients with a 'bgr' value exceeding 200 were diagnosed with chronic kidney disease (Mount Sinai, Blood Sugar Test).

The 'bu' column represents blood urea levels. This measures the amount of urea nitrogen present in the bloodstream. Urea is typically filtered out by healthy kidneys, so elevated blood urea levels can indicate impaired kidney function. According to MedlinePlus, normal blood urea nitrogen (BUN) levels are generally under 20 mg/dL. In our dataset, we found that 136 out of 148 patients with a 'bu' value greater than or equal to 50 were diagnosed with chronic kidney disease (*Bun (blood urea nitrogen): Medlineplus medical test*).

The 'sc' column refers to serum creatinine levels, which measure the amount of creatinine present in the blood. Healthy kidneys filter creatinine out of the bloodstream, so elevated serum creatinine levels typically indicate reduced kidney. In our dataset, 189 out of 192 patients with a serum creatinine level greater than 1.3 mg/dL were diagnosed with chronic kidney disease (*Understanding your lab values and other CKD health numbers*).

The 'sod' column represents serum sodium levels, which allow us to monitor fluid balance and electrolyte status in kidney patients. Abnormal sodium levels can reflect, impaired kidney function, or advanced stages of chronic kidney disease. Hyponatremia (serum sodium < 135 mEq/L) and hypernatremia (serum sodium > 146 mEq/L) are both clinically concerning. The normal reference range for serum sodium is between 135 and 146 mEq/L (Dmitrieva et al.).

The 'pot' column refers to serum potassium levels. Potassium is normally excreted by the kidneys, so elevated potassium levels (hyperkalemia) can occur when kidney function declines. The normal range for serum potassium is between 3.5 and 5.0 mEq/L. In our dataset, when combined with abnormal sodium levels, 30 out of 31 patients belonged to the ckd class (Gates).

The 'hemo' column refers to a patient's hemoglobin level, which measures the amount of oxygen-carrying protein in red blood cells. Low hemoglobin levels indicate anemia. It is estimated that more than 1 in 7 people with kidney disease have anemia. The normal hemoglobin range for adults is approximately 12–16 g/dL for both males and females. In our dataset, 172 out of 212 patients with hemoglobin values outside this normal range belong to the ckd class (Vandergriendt).

The 'pcv' column refers to packed cell volume (PCV), which measures the percentage of red blood cells in a patient's blood. PCV is a key marker for detecting anemia, a common complication of chronic kidney disease. Normal PCV ranges from 36% to 47% for adult females and 40% to 53% for adult males. For the purposes of our analysis, we considered PCV values below 36% as concerning because we do not have access to the patient's gender. In our dataset, all 144 patients with PCV values below 36% were diagnosed with chronic kidney disease (*Understanding your lab work*).

The 'wbcc' column refers to white blood cell count (WBC), which measures the number of white blood cells in a patient's bloodstream. Elevated WBC levels may indicate the presence of infection. The normal WBC range is typically between 4,000 and 11,000 cells per microliter of blood (*White Blood Count (WBC): How to read the test results*).

The 'rbcc' column represents red blood cell count (RBC), which measures the number of red blood cells circulating in the bloodstream. A low RBC count is commonly seen in patients with chronic kidney disease due to anemia. In clinical practice, an RBC count below 4.0 million cells per microliter is often considered indicative of anemia. Monitoring RBC levels is important for early detection and management of anemia in CKD patients (*High Blood Pressure and Chronic Kidney Disease*).

The 'htn' column indicates whether a patient has hypertension, a condition characterized by persistently elevated blood pressure. Hypertension and chronic kidney disease are closely related as sustained high blood pressure can damage kidney structures over time. In our dataset, every patient marked with 'htn = yes' was diagnosed with chronic kidney disease (*Hypertension in CKD: Core curriculum 2019 - American journal of kidney diseases*).

The 'dm' column indicates whether a patient has been diagnosed with diabetes mellitus. Diabetes is the leading cause of chronic kidney disease, as prolonged high blood sugar levels can damage the kidneys' filtering units over time. According to the CDC, approximately 1 in 3 adults with diabetes may also have chronic kidney disease. In our dataset, every individual with diabetes was diagnosed with chronic kidney disease (*Chronic kidney disease*).

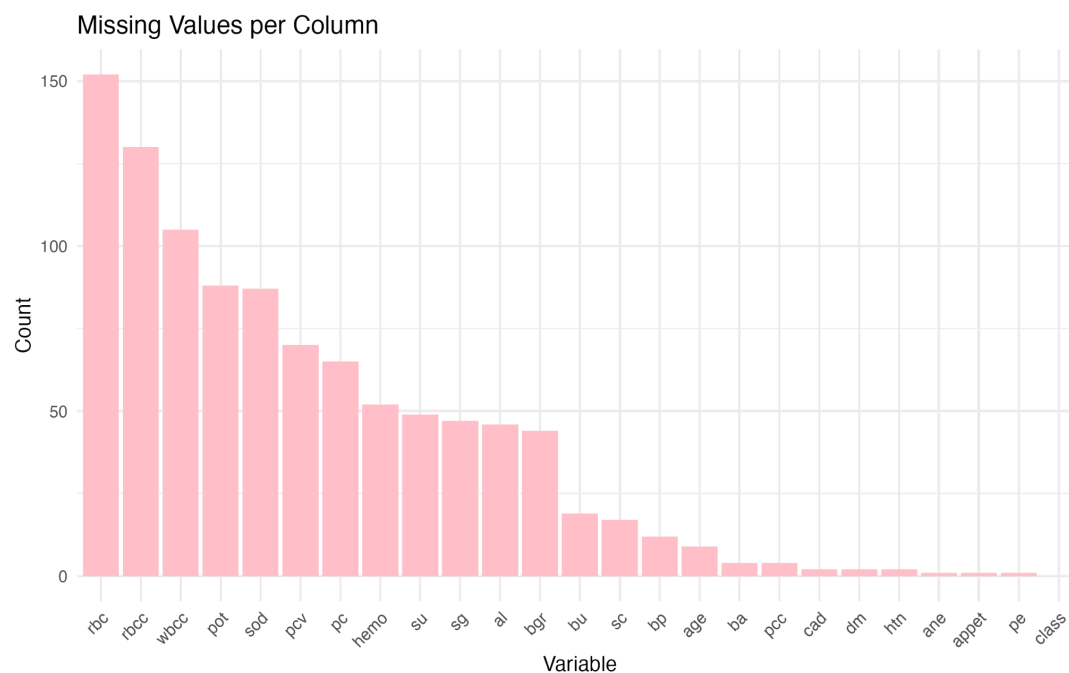
The 'cad' column refers to coronary artery disease (CAD), a condition in which narrowed or blocked arteries reduce blood flow to the heart. Individuals with chronic kidney disease are at significantly higher cardiovascular risk. The two conditions often share underlying causes such as hypertension and diabetes. (Cai et al.).

The last three predictors 'appet' (appetite), 'pe' (pedal edema), and 'ane' (anemia) represent common clinical symptoms associated with chronic kidney disease. Loss of appetite is often seen in advanced chronic kidney disease due to declining nutritional status. Pedal edema refers to swelling in the feet or ankles caused by fluid retention. Anemia indicates whether a patient has been clinically diagnosed with low red blood cell levels. These variables provide important context for assessing disease severity and patient health.

2. Methodology

Data Preprocessing

The original data file from the UCI Machine Learning Repository was provided in .arff format. To begin preprocessing, we read the data into R using the `read.table()` function. To improve readability and clarity, we renamed the column names in the dataset using the `colnames()` function. Upon inspecting the first few rows using `head(data)`, we immediately noticed the presence of missing and abnormal values. To understand the extent of missing data, we ran `colSums(is.na(data))`, which returned the number of missing values for each column. The plot below provides a visual representation of the number of missing values per predictor.



Initially, we considered removing all rows containing missing values using the `na.omit()` function. However, this approach reduced our dataset from 400 to only 158 samples which we determined would not be sufficient for our analysis and model training. Additionally, all columns contained less than 50% missing values, so removing these rows would result in significant data loss. Thus, we decided to impute missing values instead, allowing us to perform analysis on a larger sample.

In addition to missing values, the dataset contained improperly formatted data. For example, some cells included characters `'\t'` or had values like `'\t?'`, which represented missing data but were not recognized as missing. To address this, we used `lapply()` to iterate over all columns and clean character entries. We replaced all instances of `'\t?'` with `'NA'` to identify them as missing and removed any leading tab characters `'\t'` from string values which allowed for consistent formatting.

After reformatting the strings, we checked the data types of all columns. Several variables were initially stored as character values but represented either categorical or numeric data. We converted `sg`, `al`, and `su` to factors, as these variables represent discrete categories with a fixed number of levels. Additionally, we converted `pcv`, `wbcc`, and `rbcc` to numeric data types.

For imputation, we applied different techniques based on whether the variable was numerical or categorical. For numerical variables, we used predictive mean matching (`pmm`) from the `mice` package. This method fills in missing values by using existing values from similar observations. For categorical variables, we created a `get.mode()` function to fill in missing values with the most repeated value in each variable. After applying these imputation methods, all missing values in the dataset were addressed while maintaining data integrity.

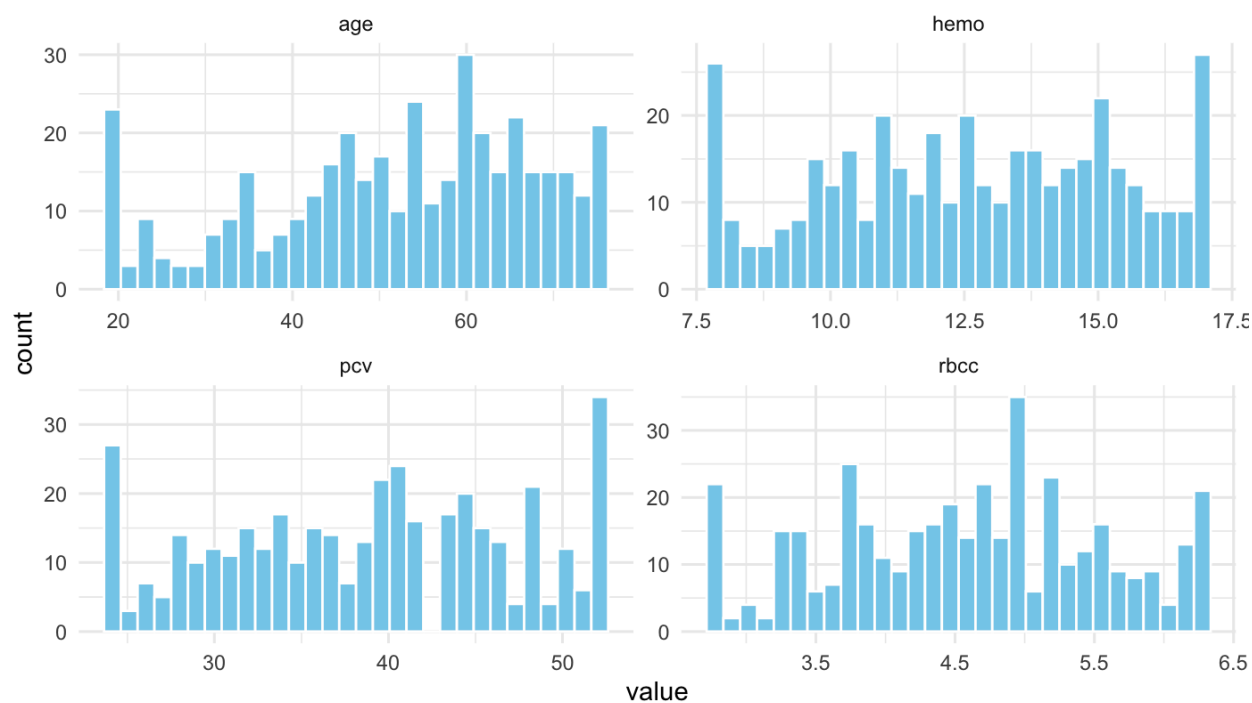
After imputation, we prepared the dataset for modeling by factorizing the appropriate variables. We converted the target variable class to binary, where '1' represented patients diagnosed with chronic kidney disease ('ckd') and '0' represented patients without chronic kidney disease ('notckd'). Additionally, we converted all remaining character variables in the dataset to factors using the `lapply()` function. This ensured that categorical variables were correctly formatted and could be properly interpreted by machine learning algorithms during classification tasks.

To identify outliers in the dataset, we created a custom `get_outliers()` function, which detected outliers in numerical variables based on the built-in `boxplot.stats()` function. Here, values falling outside of the whiskers in a boxplot were flagged as extreme values. The function then returned the columns containing outliers, and we found that all numerical variables in the dataset contained at least one outlier.

After taking a look at the outliers, we assessed the distribution of each numerical variable by calculating its skewness using the `skewness()` function from the `e1071` package. Variables with a skewness greater than 1 or less than -1 were classified as highly or negatively skewed. Variables with skewness between -1 and 1 were considered more stable. Based on this analysis, we identified `bp`, `bgr`, `bu`, `sc`, `pot`, and `wbcc` as highly skewed variables. In contrast, variables such as `age`, `hemo`, `pcv`, and `rbcc` exhibited low skewness and were treated as non-skewed.

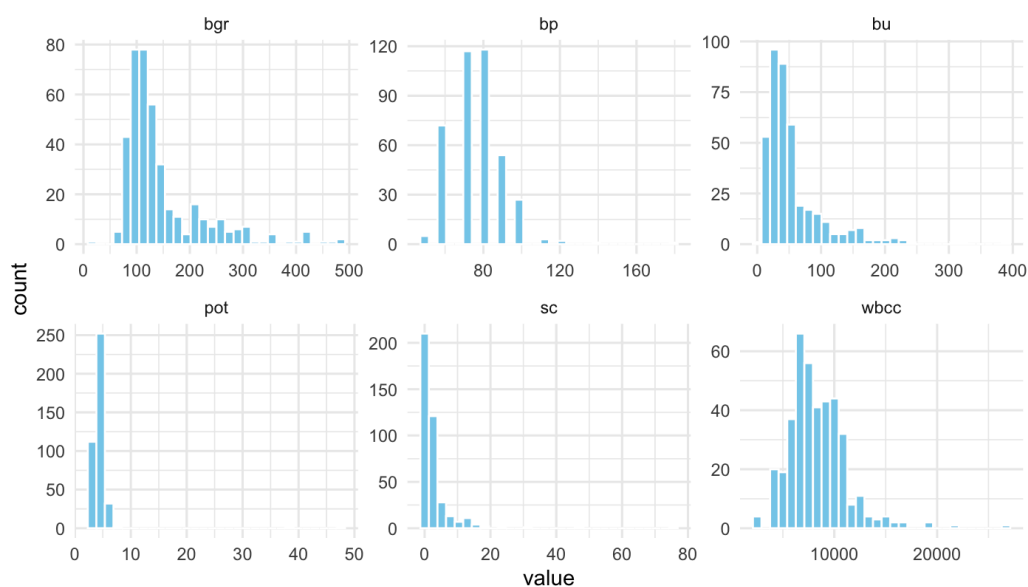
For non-highly skewed variables (`age`, `hemo`, `pcv`, `rbcc`), we applied capping to limit extreme values. We implemented a `cap_outliers()` function, which identified the 5th and 95th percentile values for each variable and replaced values falling outside this range with the respective threshold. The histograms below show the distributions of these non-skewed variables after applying capping:

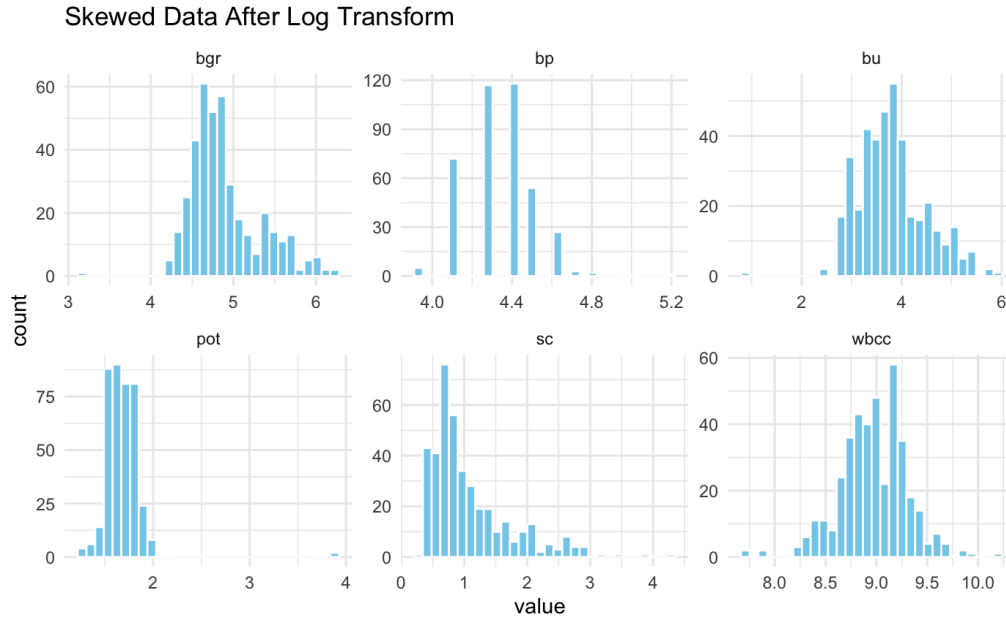
Non-skewed Data After Capping



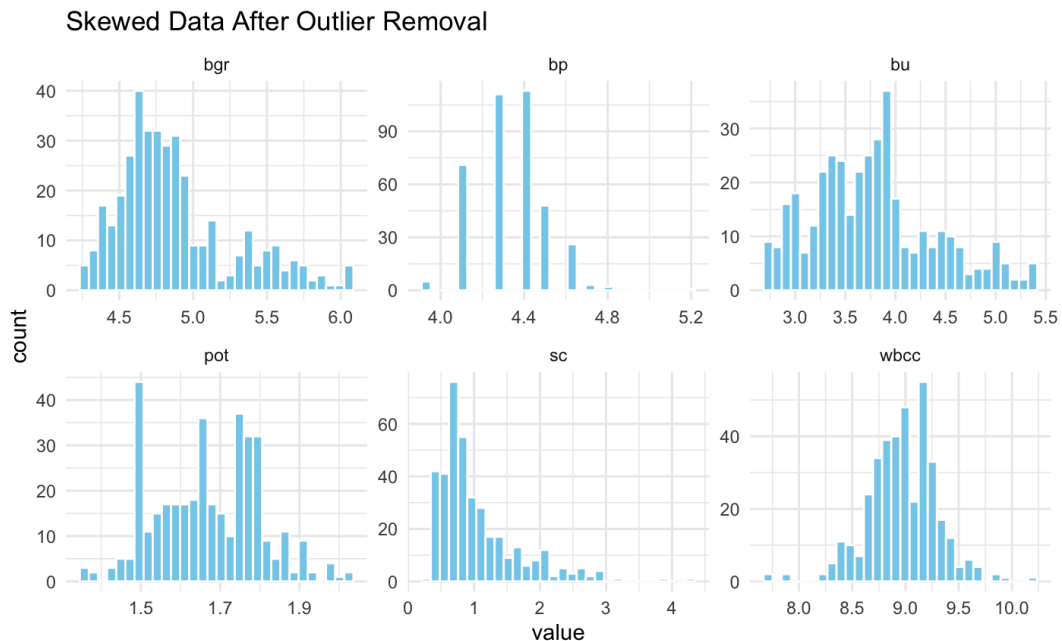
For highly skewed variables (bp, bgr, bu, sc, pot, wbcc), we applied a log transformation using the `log1p()` function to reduce skewness and help the distributions better resemble a normal, bell-shaped curve. The histograms below show the distributions of the skewed variables before and after applying the log transformation:

Skewed Data Before Log Transform



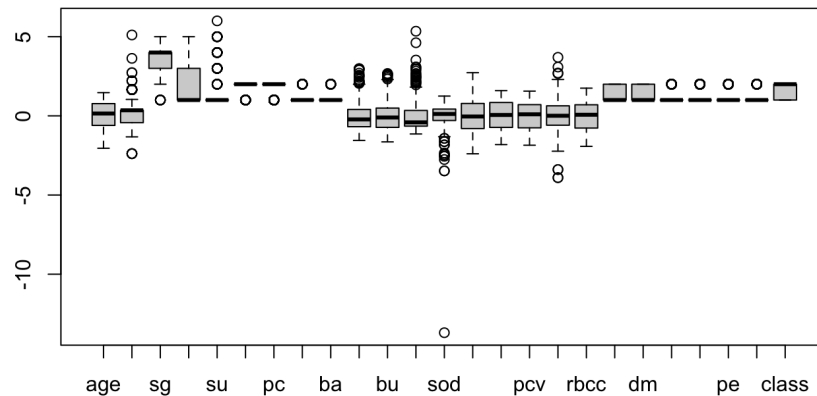


For the highly skewed variables pot, bu, and bgr, we decided to apply outlier removal after log transformation. To do so, we removed outliers by excluding values below the 1st percentile and above the 99th percentiles. After applying this outlier removal step, our final dataset contained 381 observations. We visualized the distributions post-removal to confirm that the data appeared more stable and suitable for modeling.



After removing outliers from pot, bu, and bgr, we standardized all numerical variables using the `scale()` function. This process transformed the variables to follow a standard normal

distribution with a mean of 0 and a standard deviation of 1. Additionally, we recorded the original means and standard deviations of each numeric variable using the `attr()` function. This allowed us to later reverse the scaling for interpretation purposes. The boxplot below visualizes the distributions of all variables after standardization, confirming that the numerical variables were successfully scaled and centered around zero.



To reduce dimensionality and improve model interpretability, we applied Elastic Net regularization for feature selection. Elastic Net regularization is a combination of LASSO and Ridge regression that performs both variable selection and regularization. This method is useful when dealing with highly correlated features, as it tends to retain one variable while shrinking the coefficients of the others to zero (Kassambara and Chandrakumaran *Penalized regression essentials: Ridge, Lasso & Elastic Net*). We used the `glmnet` package in R and applied 10-fold cross-validation to select the optimal tuning parameters for the model. Based on the resulting model coefficients, variables with nonzero coefficients were retained for further analysis, while variables with little to no correlation with the target variable (`class`) were removed. As a result, `rbc`, `pc`, `pcc`, `ba`, `htn`, `dm`, `cad`, `appet`, `pe`, and `ane` were removed from the dataset. The remaining variables consisted of a combination of features that were informative for predicting chronic kidney disease. The `str()` output below shows the final structure of our dataset after feature selection, outlier handling, and standardization

```
'data.frame':  381 obs. of  14 variables:
 $ age  : num  -0.228 -2.045 0.649 -0.04 0.524 ...
 $ bp   : num   0.352 -2.386 0.352 0.352 1.041 ...
 $ sg   : Factor w/ 5 levels "1.005","1.01",...: 4 4 2 2 3 2 3 3 4 2 ...
 $ al   : Factor w/ 6 levels "0","1","2","3",...: 2 5 3 3 4 1 3 4 3 4 ...
 $ su   : Factor w/ 6 levels "0","1","2","3",...: 1 1 4 1 1 1 5 1 1 1 ...
 $ bgr  : num  -0.182 -0.66 2.969 -0.514 -1.413 ...
 $ bu   : num  -0.271 -1.362 0.347 -0.787 -0.849 ...
 $ sc   : num  -0.4109 -0.7359 -0.0204 -0.27 -0.4863 ...
 $ sod  : num   0.223 0.223 -1.112 1.25 0.428 ...
 $ pot  : num  -0.4867 0.6533 0.5197 -0.0391 -1.8259 ...
 $ hemo : num   0.993 -0.543 -1.18 -0.431 -0.206 ...
 $ pcv  : num   0.586 -0.146 -1 -0.512 -0.024 ...
 $ rbcc : num   0.595 -0.455 0.175 -0.035 -0.245 ...
 $ class: Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
```

Research Question 1

Our first research question asks “Can we predict whether a patient has chronic kidney disease based on clinical and laboratory features?” To address this classification task, we implemented two supervised learning algorithms: logistic regression and random forest classification.

In statistical analysis, “logistic regression is used to predict the class (or category) of individuals based on one or multiple predictor variables. It is used to model a binary outcome, that is a variable, which can have only two possible values: 0 or 1, yes or no, diseased or non-diseased” (Kassambara et al. *Logistic Regression Essentials in R*). In our case, it is whether a patient is classified as ‘ckd’ (diagnosed with chronic kidney disease) or ‘notckd’ (not diagnosed with chronic kidney disease). To fit our logistic regression model, we first set a random seed, `set.seed(597)`, to ensure the reproducibility of our results. We then split the dataset into an 80% training set and a 20% testing set to evaluate model performance. The logistic model was fitted using the `glm()` function in R, with the formula `class ~ .` to include all available predictors. We specified the family = binomial parameter to indicate that the outcome variable was binary, either ckd or notckd.

Similarly, the random forest algorithm “is one of the most commonly used and the most powerful machine learning techniques” (Kassambara, *Bagging and Random Forest Essentials*). In our analysis, we used the `randomForest()` function from the `randomForest` package in R to fit the model. We again used `class ~ .` to include all predictors. After training the model, we used the `predict()` function to generate predictions on the testing set. To evaluate model performance, we constructed a confusion matrix by comparing the predicted class labels with the actual class labels in the test set using the `confusionMatrix()` function from the `caret` package. This allowed us to compare the accuracy, specificity, and sensitivity of the random forest model to that of the logistic regression model. Additionally, we used the `varImpPlot()` function to visualize variable importance scores. This output provided valuable insight into which variables had the greatest influence in predicting chronic kidney disease. This setup allowed us to model the probability that a patient belongs to the ckd class.

Research Question 2

To address the question, “Can we identify distinct subgroups of patients based on their clinical and laboratory features?”, we used k-means and hierarchical clustering to identify the subgroups of patients. These are common clustering algorithms for partitioning a dataset into k subgroups.

A good clustering algorithm is one that minimizes the within-cluster variation. K-means clusters are calculated using Euclidean distance. As such, only numerical variables can be used to determine the clusterings. We also have to scale the data first as variables with larger ranges will

have greater distances and dominate the calculations. We also remove outliers for this analysis as it skews our clustering.

To choose the k subgroups, we used the “elbow method.” This involves calculating the within-cluster sum of squares (WSS) for various values of k . For each k , we plot the curve of the WSS. The point where the change in WSS decreases (looks like a bend, or an ‘elbow’ in the plot) corresponds to the optimal k . Once an optimal k is determined, we use the built in `kmeans()` function to assign each data point a cluster and analyze.

An alternative approach that we investigated was hierarchical clustering. This does not require us to initialize a choice for k . From the library ‘cluster’, we use the `hclust()` function to build the model and plot a dendrogram. This algorithm works by starting with each point in its own cluster, identifying the two closest clusters and merging them, and repeats until all points are in a single cluster. This model and clustering is also analyzed and compared to the previous model. As this model can include both categorical and numerical features, this model described our data better and was selected as our most accurate model.

Research Question 3

Our third research question asked “Can chronic kidney disease be accurately predicted when key predictors such as hemoglobin appear normal?” To address this question, we used the CART (classification and regression trees) decision tree algorithm to explore feature importance and identify key variables that may be the strongest indicator of chronic kidney disease. Then, based on our research, we exclude those objects with key features that are not within ranges that are considered “normal” in the Medical field and build our models using the rest of the predictors.

3. Results

Research Question 1

Once fitting the logistic regression and random forest classification models, we used the `predict()` function to generate predicted probabilities on the test set. These probabilities represent the model’s estimated likelihood that each patient in the test set has chronic kidney disease. To evaluate the models’ classification performance, we applied a threshold of 0.5 to convert these probabilities into binary class predictions (ckd or notckd).

We then constructed a confusion matrix to summarize the number of true positives, true negatives, false positives, and false negatives. From this matrix, we extracted key performance metrics of accuracy, sensitivity, and specificity. In our case, Accuracy indicates the overall

proportion of correct predictions. Sensitivity measures how effectively the model identifies CKD patients. This is also known as the true positive rate. Specificity reflects how well the model correctly identifies non chronic kidney disease patients. This is also known as the true negative rate.

The confusion matrix on the left shows the results for the Logistic Regression model, which achieved an accuracy of 97.33%, a sensitivity of 95.65%, and a specificity of 100%. This indicates that the model was highly effective at correctly identifying patients with chronic kidney disease while perfectly classifying non-chronic kidney disease patients.

On the right, the random Forest model performed slightly better, achieving an accuracy of 98.67%, a sensitivity of 97.83%, and a specificity of 100%. The perfect specificity indicates that the model was highly reliable in ruling out chronic kidney disease in healthy patients. Additionally, the higher sensitivity suggests that the Random Forest model was more effective at correctly detecting patients with chronic kidney disease.

Both models demonstrated strong performance on this dataset. However, the Random Forest model achieved the highest accuracy and sensitivity, making it the preferred model for predicting chronic kidney disease.

Confusion Matrix and Statistics	Confusion Matrix and Statistics
Reference Prediction 0 1 0 29 2 1 0 44 Accuracy : 0.9733 95% CI : (0.907, 0.9968) No Information Rate : 0.6133 P-Value [Acc > NIR] : 1.375e-13 Kappa : 0.9445 McNemar's Test P-Value : 0.4795 Sensitivity : 0.9565 Specificity : 1.0000 Pos Pred Value : 1.0000 Neg Pred Value : 0.9355 Prevalence : 0.6133 Detection Rate : 0.5867 Detection Prevalence : 0.5867 Balanced Accuracy : 0.9783 'Positive' Class : 1	Reference Prediction 0 1 0 29 1 1 0 45 Accuracy : 0.9867 95% CI : (0.9279, 0.9997) No Information Rate : 0.6133 P-Value [Acc > NIR] : 5.768e-15 Kappa : 0.9721 McNemar's Test P-Value : 1 Sensitivity : 0.9783 Specificity : 1.0000 Pos Pred Value : 1.0000 Neg Pred Value : 0.9667 Prevalence : 0.6133 Detection Rate : 0.6000 Detection Prevalence : 0.6000 Balanced Accuracy : 0.9891 'Positive' Class : 1

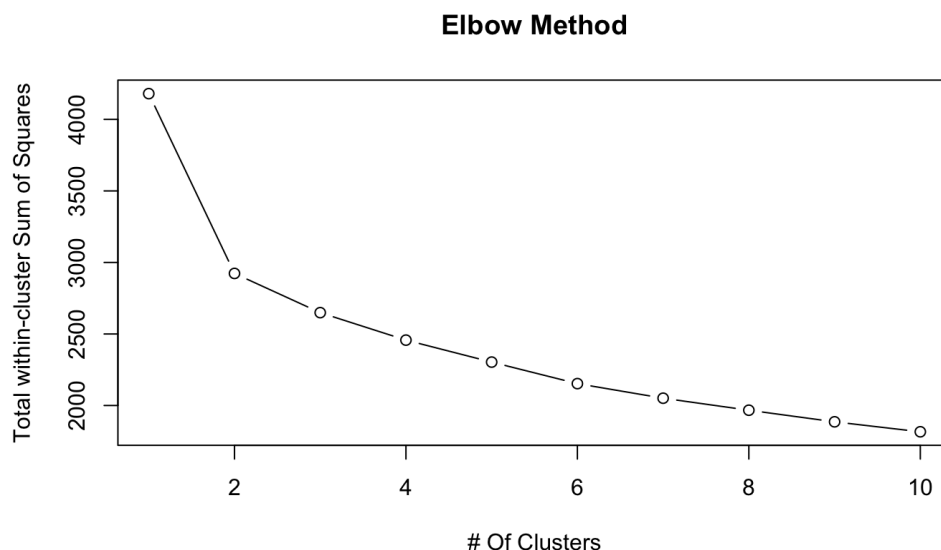
Logistic Regression Confusion Matrix

Random Forest Confusion Matrix

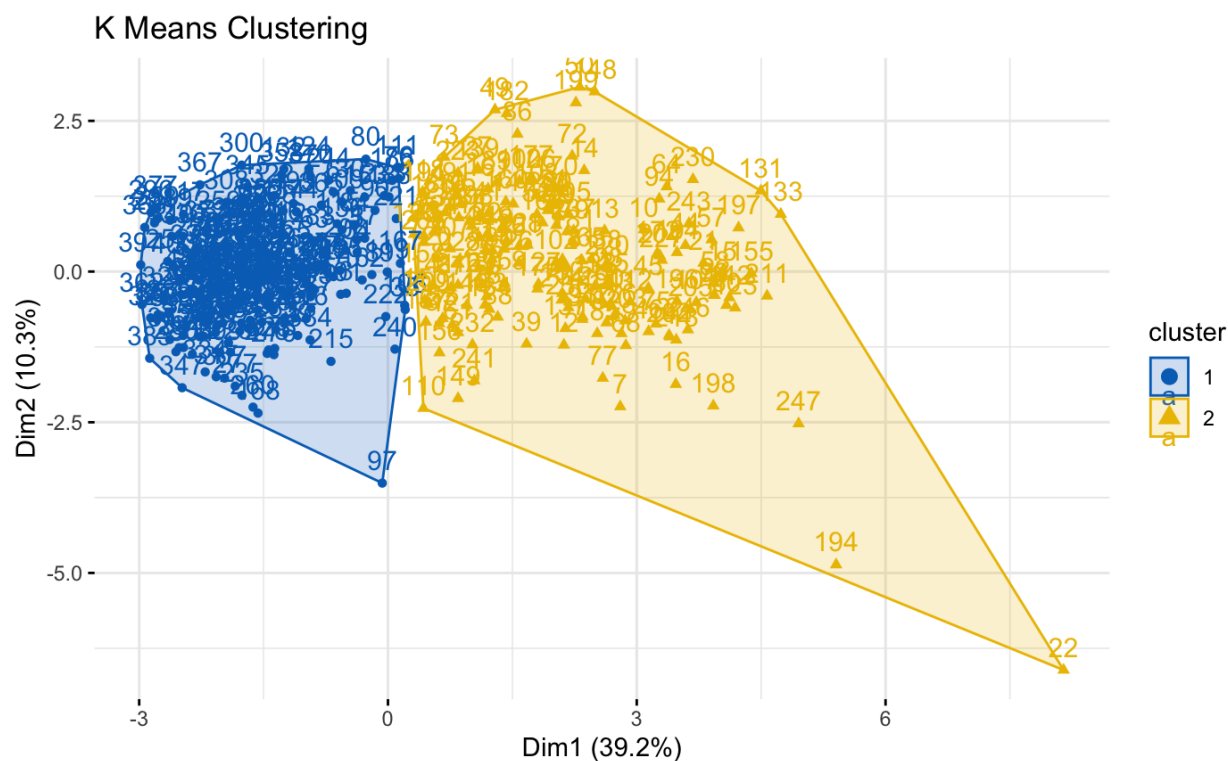
Research Question 2

We performed both K Means clustering and Hierarchical clustering algorithms to our dataset and compared results.

We first performed K-Means clustering and used the elbow method to choose $k = 2$, as shown in the figure below.



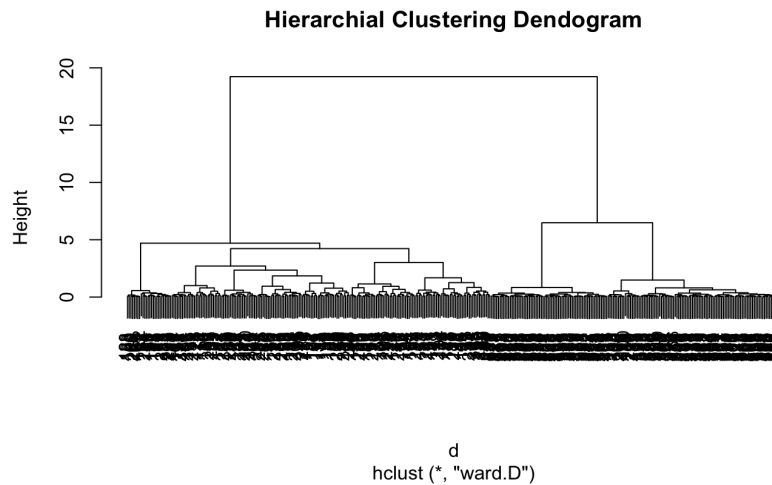
We then use the built in `kmeans()` function to assign each point to one of two clusters and plot:



Cluster	age	bp	bgr	bu	sc	sod	pot	hemo	pcv	wbcc	rbcc
1	45.82	72.2	111.05	29.93	1	140.95	4.22	14.54	44.71	8.93	5.24
2	59.51	79.76	161.09	68.62	3.56	133.61	4.47	10.33	31.75	9.03	3.82

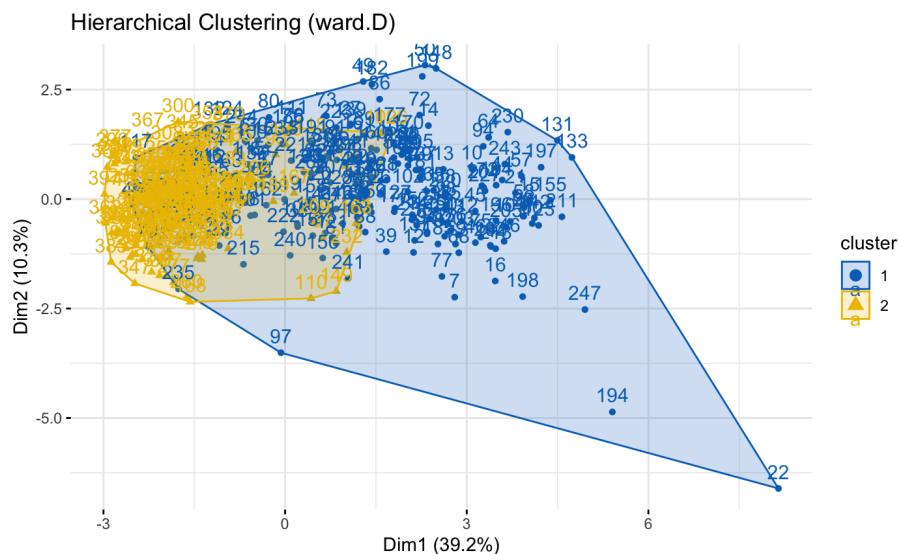
The table above shows the mean values for each variable for each group generated by the k - means clustering algorithm. Cluster 2 represents the patients with chronic kidney disease.

We also perform hierarchical clustering and generate the dendrogram to choose the number of clusters to select:



cluster	age	bp	bgr	bu	sc	sod	pot	hemo	pcv	rbcc	sg	al	su
1	47.13	71.31	111.19	31.66	1	140.93	4.27	14.84	45.49	5.31	1.02	0	0
2	55.23	78.69	147.43	54.05	2.75	135.36	4.38	11.08	34.18	4.09	1.01	0	0

The table above shows the mean values for each variable for each group generated by the hierarchical clustering algorithm. Cluster 2 represents the patients with chronic kidney disease. Below is the plot of the clusters in 2D space:



Next we generate the confusion matrices for the two clustering methods and compare.

Confusion Matrix and Statistics		Confusion Matrix and Statistics	
Reference		Reference	
Prediction	0 1	Prediction	0 1
0	148 71	0	148 21
1	0 162	1	0 212
Accuracy : 0.8136		Accuracy : 0.9449	
95% CI : (0.7709, 0.8515)		95% CI : (0.917, 0.9656)	
No Information Rate : 0.6115		No Information Rate : 0.6115	
P-Value [Acc > NIR] : < 2.2e-16		P-Value [Acc > NIR] : < 2.2e-16	
Kappa : 0.6393		Kappa : 0.8869	
McNemar's Test P-Value : < 2.2e-16		McNemar's Test P-Value : 1.275e-05	
Sensitivity : 0.6953		Sensitivity : 0.9099	
Specificity : 1.0000		Specificity : 1.0000	
Pos Pred Value : 1.0000		Pos Pred Value : 1.0000	
Neg Pred Value : 0.6758		Neg Pred Value : 0.8757	
Prevalence : 0.6115		Prevalence : 0.6115	
Detection Rate : 0.4252		Detection Rate : 0.5564	
Detection Prevalence : 0.4252		Detection Prevalence : 0.5564	
Balanced Accuracy : 0.8476		Balanced Accuracy : 0.9549	
'Positive' Class : 1		'Positive' Class : 1	

K-means Clustering Confusion Matrix

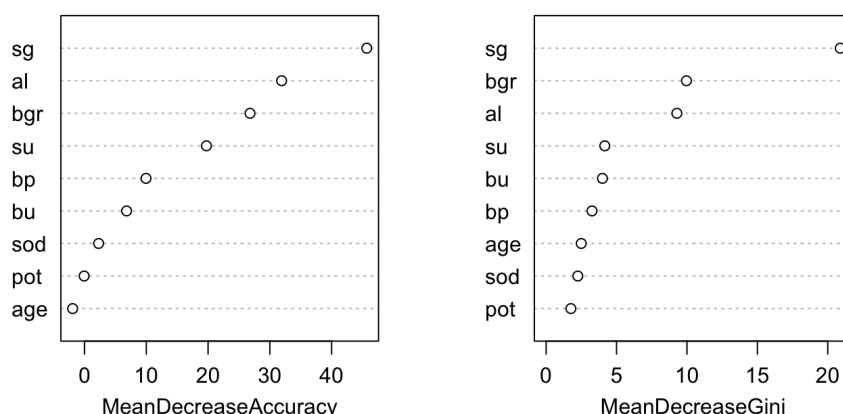
Hierarchical Clustering Confusion Matrix

The tables above show the confusion matrices for the results predicted by the k-means clustering model (left) and the hierarchical clustering model (right). As you can see, the overall accuracy of the hierarchical clustering model is higher. Both have 100% specificity, which means that neither model returns a false positive. The sensitivity of the hierarchical clustering is higher, which means this model more accurately clusters patients who have chronic kidney disease into the correct group.

Research Question 3

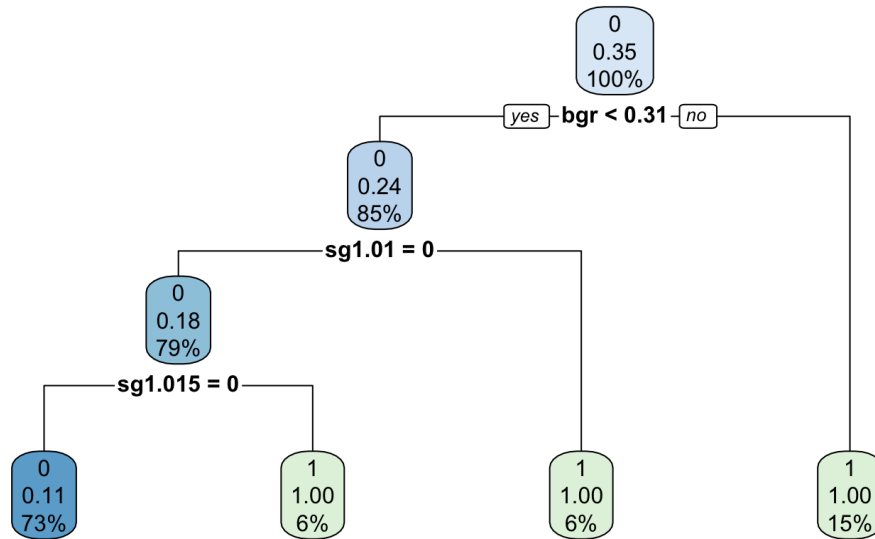
The variable importance scores from the initial exploratory CART model provided insight into which features were most influential in predicting chronic kidney disease. In particular, variables such as hemo (hemoglobin), pcv (packed cell volume), sc (serum creatinine), rbcc (Red Blood Cell Count) were ranked as highly important in the model. This suggests that abnormalities in these lab measurements may serve as early indicators of potential kidney dysfunction.

Random Forest Feature Importance for CKD



Our Random Forest model achieved an accuracy of 96.77%, with a sensitivity of 90.91% and a specificity of 100%. The 100% specificity indicates that the model is highly effective at correctly identifying non chronic kidney disease patients. This may be useful in settings where correctly ruling out chronic kidney disease in healthy patients is critical.

The CART decision tree allowed us to visualize how specific variables contributed to the classification of chronic kidney disease versus non-chronic kidney disease patients. As shown in the figure below, the most important variable for the first split was bgr (blood glucose random). Patients with lower bgr values (less than 0.31 after standardization) were more likely to not have chronic kidney disease. For patients with higher bgr values, the model then considered sg (specific gravity) values for further splits. Lower sg values were associated with non chronic kidney disease patients, while higher sg values or abnormal patterns in sg increased the likelihood of chronic kidney disease classification.



Similar to the random forest model, the simpler tuned CART model built using the `train()` method also performs well in identifying healthy patients while achieving reasonable sensitivity in diagnosing chronic kidney. This suggests that even when the strongest predictors appear normal, early signs of CKD can still be detected from secondary features.

Our CART decision tree model achieved an accuracy of 90.32%, with a sensitivity of 72.73% and a specificity of 100%. The 100% specificity indicates that the model is highly effective at correctly identifying non chronic kidney disease patients. However, the sensitivity of 72.73% suggests that the model was less effective in identifying chronic kidney disease patients compared to the random forest model.

4. Conclusions and Discussion

While we were able to apply various data preprocessing techniques and build models that performed well on our dataset, the results of our analysis were not entirely aligned with findings from existing scientific research.

For example, it is well-documented that hypertension is one of the leading causes of chronic kidney disease. However, we removed `htn` (hypertension) as a predictor during our feature selection process using elastic net regularization because `htn` showed little statistical association with the target variable (class). This discrepancy may have been due to the limited sample size as we only had 381 observations after data cleaning and preprocessing within the dataset. Similarly, other medically significant variables such as `dm` (diabetes mellitus) and `cad` (coronary artery

disease) which are both known risk factors for chronic kidney disease, were also excluded because of their low correlation to the target variable during feature selection.

We believe that our analysis could have been improved and more scientifically aligned if we had access to a larger dataset with more detailed predictors. For instance, the dataset only included diastolic blood pressure (bp), whereas including both systolic and diastolic blood pressure measurements would have provided a more complete understanding of a patient's cardiovascular health. Additionally, the dataset did not account for patient sex or gender, which may influence certain clinical measurements and risk factors. Variables such as normal hemoglobin levels and creatinine levels often differ between male and female patients. Other useful predictors that could have enhanced the analysis include medication history, family history of kidney disease, duration of hypertension or diabetes, and lifestyle factors.

Expanding the dataset to include additional features and increasing the sample size would likely produce more accurate and relevant models for predicting chronic kidney disease.

One last consideration to note is that much of our scientific research and background information was based on studies conducted in the United States. However, the dataset we used was collected from patients in Tamil Nadu, India. While this may not have a significant effect on the risk factors for chronic kidney disease, as adults should generally exhibit similar patterns of disease progression, there may still be regional differences in healthcare access, diagnostic practices, or data collection methods that could have influenced our results.

5. References

“Aging and Kidney Disease.” National Kidney Foundation, 12 Aug. 2024,

www.kidney.org/aging-and-kidney-disease#:~:text=Kidney%20disease%20can%20develop%20at,believed%20to%20have%20kidney%20disease

“Blood in Urine (Hematuria).” Mayo Clinic, Mayo Foundation for Medical Education and

Research, 7 Jan. 2023, [www.mayoclinic.org/diseases-conditions/blood-in-urine/](https://www.mayoclinic.org/diseases-conditions/blood-in-urine/symptoms-causes/syc-20353432#:~:text=This%20condition%20happens%20when%20the,the%20body%2C%20called%20the%20urethra)

[symptoms-causes/syc-20353432#:~:text=This%20condition%20happens%20when%20the,the%20body%2C%20called%20the%20urethra](https://www.mayoclinic.org/diseases-conditions/blood-in-urine/symptoms-causes/syc-20353432#:~:text=This%20condition%20happens%20when%20the,the%20body%2C%20called%20the%20urethra)

“Blood Sugar Test - Blood.” Mount Sinai Health System, www.mountsinai.org/health-library/

tests/blood-sugar-test-blood#:~:text=In%202022%2C%20the%20US%20Preventive,mm
ol/L)%20or%20lower. Accessed 14 Apr. 2025

Boehmke, Brad, and Brandon M Greenwell. *Hands-on Machine Learning with R*. CRC Press, 7
Nov. 2019

“Bun (Blood Urea Nitrogen): Medlineplus Medical Test.” MedlinePlus, U.S. National Library of
Medicine, medlineplus.gov/lab-tests/bun-blood-urea-nitrogen/#:~:text=But%
20higher%2Dthan%2Dnormal%20BUN,may%20not%20have%20any%20symptoms.
Accessed 14 Apr. 2025

Cai, Qiangjun, et al. “Coronary Artery Disease in Patients with Chronic Kidney Disease: A
Clinical Update.” Current Cardiology Reviews, U.S. National Library of Medicine, Nov.
2013, pmc.ncbi.nlm.nih.gov/articles/PMC3941098/#:~:text=Chronic%20kidney%
20disease%20 (CKD)%20is,strategy%20for%20clinically%20stable%20CAD

“Chronic Kidney Disease (CKD) - NIDDK.” National Institute of Diabetes and Digestive and
Kidney Diseases, U.S. Department of Health and Human Services, www.niddk.nih.gov
/health-information/kidney-disease/chronic-kidney-disease-ckd. Accessed 8 Apr. 2025

“Chronic Kidney Disease.” Centers for Disease Control and Prevention, Centers for Disease
Control and Prevention, www.cdc.gov/diabetes/diabetes-complications/ diabetes-
and-chronic-kidney-disease.html. Accessed 14 Apr. 2025

“Chronic Kidney Disease.” Mayo Clinic, Mayo Foundation for Medical Education and Research,
6 Sept. 2023, www.mayoclinic.org/diseases-conditions/chronic-kidney-disease/
symptoms-causes/syc-20354521

“Chronic Kidney Disease.” UCI Machine Learning Repository, archive.ics.uci.edu/
dataset/336/chronic+kidney+disease. Accessed 13 Apr. 2025

Dicu-Andreescu, Ioana, et al. “Chronic Kidney Disease, Urinary Tract Infections and Antibiotic

- Nephrotoxicity: Are There Any Relationships?” *Medicina* (Kaunas, Lithuania), U.S. National Library of Medicine, 27 Dec. 2022, [pmc.ncbi.nlm.nih.gov/articles/PMC9862113/#:~:text=CKD%20represents%20a%20risk%20factor,category%20of%20patients%20%5B6%5D](https://pubmed.ncbi.nlm.nih.gov/articles/PMC9862113/#:~:text=CKD%20represents%20a%20risk%20factor,category%20of%20patients%20%5B6%5D)
- Dmitrieva, Natalia I, et al. “Middle Age Serum Sodium Levels in the Upper Part of Normal Range and Risk of Heart Failure.” *European Heart Journal*, U.S. National Library of Medicine, 14 Sept. 2022, [pmc.ncbi.nlm.nih.gov/articles/PMC10263272/#:~:text=8,glucose%20being%20the%20main%20contributors](https://pubmed.ncbi.nlm.nih.gov/articles/PMC10263272/#:~:text=8,glucose%20being%20the%20main%20contributors)
- “High Blood Pressure and Chronic Kidney Disease.” National Kidney Foundation, 12 Aug. 2024, www.kidney.org/high-blood-pressure-and-chronic-kidney-disease
- Hypertension in CKD: Core Curriculum 2019 - *American Journal of Kidney Diseases*, [www.ajkd.org/article/S0272-6386\(19\)30094-0/fulltext](https://www.ajkd.org/article/S0272-6386(19)30094-0/fulltext). Accessed 14 Apr. 2025
- Gates B Colbert, MD. “Potassium.” Reference Range, Interpretation, Collection and Panels, Medscape, 27 Feb. 2024, emedicine.medscape.com/article/2054364-overview?form=fpf
- “Glycosuria: What It Means & When You Need Treatment.” Cleveland Clinic, 19 Mar. 2025, my.clevelandclinic.org/health/diseases/glycosuria
- Kassambara. “Bagging and Random Forest Essentials.” *STHDA*, 10 Mar. 2018, www.sthda.com/english/articles/35-statistical-machine-learning-essentials/140-bagging-and-random-forest-essentials#computing-random-forest-classifier
- Kassambara, et al. “Logistic Regression Essentials in R.” *STHDA*, 11 Mar. 2018, www.sthda.com/english/articles/36-classification-methods-essentials/151-logistic-regression-essentials-in-r/
- Kassambara, and Chandrakumaran. “Penalized Regression Essentials: Ridge, Lasso & Elastic

- Net.” STHDA, 11 Mar. 2018, www.sthda.com/english/articles/37-model-selection-essentials-in-r/153-penalized-regression-essentials-ridge-lasso-elastic-net/
- “Kidney Failure Risk Factor: Serum Albumin.” National Kidney Foundation, 12 Aug. 2024, www.kidney.org/kidney-failure-risk-factor-serum-albumin
- Nall, Rachel. “Urine Specific Gravity Test: Overview, Tests, and Procedure.” Healthline, Healthline Media, 18 Apr. 2023, www.healthline.com/health/urine-specific-gravity#results
- PathLabs, Team Dr Lal. “Pus Cells in Urine: Causes & Diagnosis - Dr Lal PathLabs Blog.” Dr Lal PathLabs, 25 Mar. 2025, www.lalpathlabs.com/blog/pus-cells-in-urine/#:~:text=Why%20are%20Pus%20Cells%20Present,in%20elevated%20pus%20cell%20levels
- “Understanding Your Lab Values and Other CKD Health Numbers.” National Kidney Foundation, 13 Apr. 2025, www.kidney.org/kidney-topics/understanding-your-lab-values-and-other-ckd-health-numbers#:~:text=It%20is%20removed%20from%20you, and/or%20chronic%20kidney%20disease
- “Understanding Your Labwork.” DaVita, www.davita.com/education/kidney-disease/symptoms/understanding-your-lab-work#:~:text=Purpose:%20To%20determine%20the%20percentage,is%2033%20to%2036%20percent. Accessed 14 Apr. 2025
- Vandergriendt, Carly. “What Is Hemoglobin? Normal Levels and Potential Problems.” Healthline, Healthline Media, 12 July 2023, www.healthline.com/health/what-is-hemoglobin#blood-test
- “White Blood Count (WBC): How to Read the Test Results.” Ada, ada.com/white-blood-cell-count/. Accessed 14 Apr. 2025

6. Appendix

1. [Code in .rmd file](#)
2. [Code in .pdf file](#)
3. [Data in .arff file](#) (read by running our code)
4. [Presentation slides](#)
5. [Project folder](#)